

Use-Case Flow Specification

UC-01: Analyse Query Execution Plan | Database Query Performance Profiler

Primary Actor	Database Administrator
Supporting Actor	Backend Lead
Trigger	A Database Administrator selects an ingested slow query and requests plan analysis.
Preconditions	The user is authenticated as a Database Administrator; a slow-query record and a valid PostgreSQL or MySQL EXPLAIN plan are available.
Postconditions	The analysis result, detected issues, and any index recommendation are stored and visible in the query profile.

Main Success Scenario

- 1 The Database Administrator opens a slow-query record from the ranked query list.
- 2 The system displays the normalized query details and the associated execution plan.
- 3 The Database Administrator selects Analyse Plan.
- 4 The system validates the plan format and parses its operators, predicates, estimated rows, and costs.
- 5 The system identifies sequential scans and evaluates whether filtered or join columns lack a suitable index.
- 6 The system calculates a recommendation confidence level and prepares the proposed index definition.
- 7 The system displays the detected issue, supporting plan evidence, proposed index, and confidence level.
- 8 The Database Administrator reviews the result and selects Accept, Dismiss, or Annotate.
- 9 The system records the decision, user, timestamp, and optional annotation in the query profile.
- 10 The system confirms that the analysis and decision have been saved.

Alternate Flow A1 - Unsupported or Invalid Plan

- At step 4, if the plan cannot be parsed or comes from an unsupported format, the system marks the analysis as Failed and displays the validation error.
- The system retains the original query record and writes an audit event with the error details.
- The Database Administrator may upload a corrected supported plan or exit without creating a recommendation.